Sidratul Muntaha

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Summary

I'm a curious and driven 4th-year CSE student at CUET who genuinely enjoys solving problems - whether it's untangling logic in code or figuring out how to make systems work better. So far, I've solved 220+ problems across platforms like Codeforces, UVA, and AtCoder, and also completed structured problem sets on CodingBat, sharpening my skills in Python fundamentals.

I love the process of breaking down complex ideas into clear, efficient solutions. This mindset flows into my development work too - I've built and contributed to several full-stack projects using React, Django, and SQL, always focusing on writing clean, functional code.

Alongside development, I have a growing interest in research and applied problem-solving, particularly in data-driven and intelligent systems. I'm currently exploring AI and Machine Learning, motivated by their potential to create meaningful, real-world impact. Beyond engineering, I'm also interested in design psychology - especially how thoughtful UX decisions influence user behavior and overall experience.

Education

Examination	Major	Institute	Year	GPA
Undergraduate	CSE	Chittagong University of Engineering & Technology (CUET)	2023–2026	3.72 / 4.00 (Till Level–3, Term–2)
HSC	Science	Govt. Hazi Muhammad Mohsin College	2019–2021	5.00 / 5.00
SSC	Science	CDA Public School & College	2009–2018	5.00 / 5.00

Achievements

- Ranked **1st Runner-Up** in the Cyber Defence Innovation Challenge, outperforming 70+ teams in a national-level competition, organized by UNDP Bangladesh and The Daily Star [Nov 2024]
- Awarded **Champion** in Poster Presentation for impactful design and presentation on cyber awareness, IEEE Computer Society CUET [Feb 2024]
- Achieved **6th place internationally** in the **CHiPSAL 2026 Shared Task** at *LREC-COLING 2026* by developing a **Late-Fusion Multimodal Stacking** framework for Nepali meme sentiment analysis [2026]
- Selected as a **Top 11 Finalist among 85 teams** in the Triscend STEM-Based National Case Competition, organized by IEEE CUET WIE Affinity Group Student Branch [Jan 2026]
- Selected as a **Top 10 Finalist** in the National Seerah Fest 1445 Case Competition, presenting an analytical solution to contemporary social issues from an Islamic perspective [2024]
- Awarded **Best Student of the Year** for academic excellence, leadership, and extracurricular achievements, CDA Public School & College [Sep 2019]
- Secured **Champion** title in BTV Inter-School Debate Competition, demonstrating strong critical

thinking and persuasive communication skills [Jan 2018]

Publications

- **A Machine Learning Approach for Multi-Class Global Weather Classification with Mutual-Information–Driven Features**. Published in *2025 WIECON-ECE*, IEEE.

  DOI: 10.1109/WIECON-ECE69386.2025.11526039

  Developed a leakage-safe and imbalance-aware machine learning pipeline for multi-class weather classification using the **World Weather Repository dataset containing 90,000+ global weather records and 40+ meteorological features**. Evaluated **10+ traditional and ensemble learning models** with mutual information–based feature selection, where **XGBoost achieved 92.8% accuracy and 92.6% F1-score**. Demonstrated the effectiveness of ensemble learning for reliable weather classification in risk-sensitive domains such as **disaster management and aviation**.

- **Late-Fusion Multimodal Stacking for Nepali Meme Sentiment Analysis**. Accepted at the *CHiPSAL 2026 Shared Task* at *LREC-COLING 2026*.

  Proposed a **late-fusion multimodal stacking framework** combining textual, semantic, and visual representations for sentiment analysis of Nepali text-embedded memes in a low-resource NLP setting. The system achieved a **Macro F1-score of 0.5045** on the official benchmark test set and secured **6th place internationally** in the shared task leaderboard, outperforming numerous competing systems through effective multimodal fusion and cross-validated ensemble learning.

Experience

- Serving as a **Junior Developer** at **Climate Intelligent Technologies (CIT)**, the tech team of Green Lead, contributing to climate-tech and sustainability-focused digital solutions through software development, technical implementation, and collaborative innovation projects [Apr 2026 – Present]

Skills

Technical Skills

- Programming: C++, Python
- Web Development: HTML, CSS, JavaScript,Django, NextJS, React
- Database Management: SQL
- Core CS: Data Structures & Algorithms, Problem-Solving
- Project Management, Research & Documentation

Soft Skills

- Strategic Problem-Solving & Quick Thinking
- Team Collaboration & Leadership
- Effective Public Speaking & Presentations
- Design Thinking & Communication

Projects

Development

One-Day Job Board–A Micro-Gig Platform for CUET Students —*NextJS, TailwindCSS* *Jan 2026*

Academic Project (CSE-300, CSE-356) Project Link

- Developed a web platform addressing flexible income and time-bound work challenges for university students.

- Designed a responsive, modern UI using Tailwind CSS and implemented efficient client-side routing.

Libra Jewelry – E-commerce Web Application —*Django, SQLite, HTML/CSS, Bootstrap* *July 2025*
Academic Client-Based Project (CSE-326), USA Project Link

- Built a fully functional jewelry e-commerce platform with user authentication, product browsing, and cart management.
- Integrated secure SSLCommerz payment gateway and handled backend logic using Django templates.

Hall Room Allocation System — *Bash, Linux CLI* *July 2025*
Academic Project Project Link

- Developed a terminal-based hostel room allocation management system using Bash scripting and text-file storage.
- Implemented separate student and provost panels with authentication, room management, and allocation approval workflows.
- Designed a lightweight file-based database system for handling room, student, and allocation records.

Mentor–Mentee Scheduling System—*React, SQL* *Feb 2025*
Academic Project (CSE-252, CSE-202) Project Link

- Developed a scheduling system to match mentors and mentees based on preferences and availability.
- Worked on frontend development using React and database design using SQL.

Machine Learning

Multi-Class Weather Classification Using Machine Learning — *Python, Scikit-Learn, XGBoost, CatBoost* *Dec 2025*
IEEE Conference Research Project Project Link

- Developed a leakage-safe machine learning pipeline for multi-class weather classification using 90,000+ global weather records.
- Applied mutual-information feature selection, preprocessing, and feature engineering to improve model performance.
- Evaluated 10+ traditional and ensemble models; XGBoost achieved 92.8% accuracy and 0.991 ROC-AUC.

Diabetes Risk Classification Using Machine Learning — *Python, Scikit-learn* *May 2026*
Academic Project Project Link

- Built a multi-class diabetes risk prediction system using 6,000 patient records and 17 clinical features.
- Implemented and compared 9 supervised learning algorithms including SVM, Random Forest, Gradient Boosting, and Neural Networks.
- Developed ensemble models using Voting and Stacking techniques, achieving over 91% classification accuracy.

Maternal Health Risk Prediction System *Jan 2026*
Academic Project (CSE-312) Project Link

- Built a multi-class classification system to predict maternal health risk levels.
- Improved performance using ensemble methods and hyperparameter tuning.
- Evaluated and visualized baseline vs tuned models using standard ML metrics.

Rainfall Prediction using Machine Learning *Sep 2025*
Self Driven Project Project Link

- Developed a rainfall prediction model using real-world weather datasets.

- Implemented and compared Logistic Regression, Decision Tree, Random Forest, SVM, and KNN.
- Evaluated models using cross-validation, learning curves, and performance metrics.

Data Structures & Algorithms

DSA-Based GUI Applications —*Python, Tkinter* *Jan 2025*
Self Driven Projects Project Link

- Developed GUI-based applications including Cash Flow Minimizer and Snake Game.
- Applied Greedy Algorithms, Graphs, Heaps, Arrays, and Event-Driven Programming concepts.
- Focused on visualization and beginner-friendly interaction using Tkinter.

Design & Creative

Graphic Design Contributions —*Adobe Illustrator, Canva* *2025*
Volunteer & Design Submissions

- Designed Eid Card for CUET Islamic Ilm Seeker Society.
- Created logo design submission for CUET CSE Fest 2025.
- Designed a calendar for IEEE CUET Student Branch.

Certifications & Courses

- **Supervised Learning with scikit-learn** — DataCamp
- **Understanding Machine Learning** — DataCamp
- **Python Certification** — Kaggle
- **Phase 0: A Deep Introduction to Programming** — YouKn0wWho Academy
- **ChatGPT Prompt Engineering for Developers** — DeepLearning.AI
- **Spoken English Course** — Speakers Council Ltd. EduStation

References

Ashim Dey — Assistant Professor (CSE), CUET
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